Device Management System

Complete User Manual

WebSocket SDK — Django Web Application

Version	**2.0**
Platform	**Django 4.2 • Python 3.11+**
Devices	**TC5000 / M50 / M91 & compatible**
Protocol	**WebSocket XML SDK**

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1. System Overview & Architecture

The **Device Management System** is a Django-based web application that communicates with biometric access-control devices (fingerprint readers, face-recognition terminals, card readers) over a persistent WebSocket connection managed by the **Device Broker** service.

Architecture Diagram

System Architecture

```
┌──────────────────────────────────────────────────────────────────┐
│   BIOMETRIC DEVICES (TC5000 / M50 / M91)                          │
│   [Device A] [Device B] [Device C] ... [Device N]                 │
└──────────────────────────┬───────────────────────────────────────┘
                           │ WebSocket (XML SDK)
┌──────────────────────────────────────────────────────────────────┐
│   DEVICE BROKER                                                   │
│   • Register/Login handshake                                      │
│   • Command routing (send & receive XML)                          │
│   • Push log events to Django via HTTP                            │
└──────────────────────────┬───────────────────────────────────────┘
                           │ Unix socket / TCP
┌──────────────────────────────────────────────────────────────────┐
│   DJANGO WEB APPLICATION                                          │
│   • UI: Zones, Devices, Employees, Logs, Sync, API Keys           │
│   • REST API: /api/v1/ (X-API-Key auth)                           │
│   • Database: SQLite / PostgreSQL                                 │
└──────────────────────────────────────────────────────────────────┘
                           │ HTTP/HTTPS
┌──────────────────────────────────────────────────────────────────┐
│   EXTERNAL SOFTWARE (HR, Payroll, ERP) via REST API               │
└──────────────────────────────────────────────────────────────────┘
```

Starting the System

1. **Start the Device Broker:** `python -m devicebroker`

2. **Start the Django web server:** `python manage.py runserver 0.0.0.0:8000`

3. **Open the web UI in a browser:** `http://<server-ip>:8000/`

i Tip: Use `start_all.py` or `start_all.sh` in the root of the project to launch both services at once.

2. Device Management

> **Requirement 1 & 2:** Device Management — view all connected devices, their type, serial number, and status in real time.

The **Online Devices** page shows every biometric terminal currently connected to the system via WebSocket. Devices appear automatically when they connect and are removed when they disconnect.

How to Access

1. Open the web application in your browser.
2. Click **Online Devices** in the top navigation bar.
3. The table shows all currently connected devices with their serial number (Device ID), connection slot, type, and model name.

Home | ▶ Online Devices | Manage ▼ | Logs ▼ | API Keys

Online Devices — Live View

Device ID (Serial)	Connection ID	Terminal Type	Product Name	
TC5000-A1B2C3	0	Fingerprint+Face	TC5000	🔗 Open
M50-001122	1	Face Recognition	M50-Pro	🔗 Open
M91-AABBCC	2	Card+Fingerprint	M91 DE	🔗 Open

Green row = device is online. Click 'Open' to access all controls for that device.

Device Control Panel

Clicking **Open** for any device navigates to the full **Device Control Panel** which exposes all remote operations: user management, biometric data, logs, settings, sync, restart, and more.

Device Control Panel — TC5000-A1B2C3

Device ID	TC5000-A1B2C3
Terminal Type	Fingerprint+Face
Product Name	TC5000

- ► Device Status / Info / Misc Controls
- ► Ethernet Settings | Wi-Fi Settings | NTP Settings

- ► User Management | Face Data | Fingerprints | Card/QR/Password
- ► Remote Enroll | Enroll Face by Photo

- ► Access Timezone Settings | Lock Control
- ► Attendance Logs | Clear Data
- ► Bulk Download Logs | Restart Device

3. How to Set Up a Device

> **Requirement 2:** Procedure for connecting a physical device to the software system.

Prerequisites

- Device must be on the same network (LAN / VPN) as the server, or reachable over the internet.
- The Device Broker and Django web application must be running.

Step-by-Step Setup

1 Configure the Server URL on the Device

On the physical device, go to **Settings** → **Network** → **Server URL** (or use the **Server URL Settings** page in the web UI under Device → Server URL Settings) and enter: `ws://<server-ip>:8765`

2 Device Connects and Registers

The device initiates a WebSocket connection to the broker. The broker sends a Registration request to Django, which creates a **DeviceRegistry** record automatically and issues a unique token.

3 Device Logs In

The device sends its serial number and token. Django validates the token. On success, the device appears in the **Online Devices** list.

4 Verify in Device Registry

Navigate to **Manage** → **Device Registry**. The new device will have Online status and a `last_seen` timestamp.

Device Registration Flow	
Step 1	Device sends: Register (SN=TC5000-A1B2C3)
Step 2	Server responds: token=b3de3cba5801…
Step 3	Device sends: Login (SN=TC5000-A1B2C3, token=b3de3cba…)
Step 4	Server responds: OK — device is now online
Step 5	Django creates/updates DeviceRegistry record
Step 6	Connection event logged: Connected at 2026-04-23 10:15:00

Tokens are persisted in the DeviceRegistry table. A device that reconnects uses the same token.

✓ **Verification:** After setup, the device serial number appears in **Online Devices** and in **Device Registry** with status Online and a recent Last Seen time.

4. Add a Device to the Registry

> **Requirement 3:** Pre-register or manually add a device so it can be given a name, assigned to a zone, and configured before it connects.

Devices auto-register on first connection, but you can also pre-add them manually to assign a friendly name, location, and zone ahead of time.

How to Add a Device Manually

1 Navigate to Device Registry

Click **Manage** → **Device Registry** in the top navigation bar.

2 Click "+ Add Device"

Click the blue **+ Add Device** button in the top-right corner.

3 Fill in the Device Details

Add Device Form

Serial Number *	TC5000-ENTRANCE-01 ← exact SN printed on device
Friendly Name	Main Entrance Reader
Location	Building A, Ground Floor
Zone	[Ground Floor Zone ▼]
Active	☑ Yes (device can log in)
Interlock Enabled	☐ (enable for interlock feature)

Save Cancel

Serial Number must exactly match the SN printed on the physical device label.

4 Click Save

The device record is created. When the physical device connects, it will use this record for authentication and inherit the zone and name settings.

Device Registry — After Adding

Serial Number	Name	Zone	Location	Status	Interlock	Last Seen
TC5000-ENTRANCE-01	Main Entrance Reader	Ground Floor	Building A, GF	🟡 Offline	OFF	—
M50-001122	Cafeteria Door	1st Floor	Building B	🟢 Online	ON	2026-04-23 10:15

5. Assign a Device to a Zone / Area

Requirement 4: Organise devices into physical zones (areas) for zone-wise sync, reporting, and interlock grouping.

Step 1 — Create a Zone

1. **Go to Manage → Zones / Areas**
2. **Click "+ New Zone"**

Create Zone Form

Zone Name *	Ground Floor
Description	All entrance and exit readers on ground floor

Save Cancel

3. **Click Save**

   The zone is created and appears in the Zones list.

Step 2 — Assign Devices to the Zone

1. **Go to Manage → Device Registry**
2. **Click Edit next to the device you want to assign**
3. **Change the Zone dropdown to the desired zone and click Save**

Zones List

Zone Name	Description	Device Count	Actions
Ground Floor	All entrance/exit on ground floor	3 devices	Edit | Sync | Delete
1st Floor	Office area readers	5 devices	Edit | Sync | Delete
Parking	Parking barrier readers	2 devices	Edit | Sync | Delete

The 'Sync' button triggers a bidirectional sync of all online devices in that zone.

i Tip: You can trigger a zone-wide user sync (face, fingerprint, card data) by clicking the **Sync** button next to any zone. See Section 8 for details.

6. Clear Data from a Device

Requirement 5: Remotely erase user data, attendance logs, management logs, or all data from a selected device.

How to Clear Device Data

1. **Open the device from Online Devices → Open**
2. **Scroll to Maintenance → click Clear Data**
3. **Choose what to clear and click the corresponding button**

Clear Data Page — TC5000-A1B2C3

⚠ Warning	These operations are irreversible on the device.
Clear User Enrollment Data	Removes all user profiles + biometric templates
Take Off Managers	Demotes all manager/admin privileges
Clear Attendance Logs	Deletes all time-attendance records on device
Clear Management Logs	Deletes all admin action logs on device
Clear All Data	Wipes everything (users + all logs)

Clear User Data | Take Off Managers | Clear Attendance Logs | Clear Mgmt Logs | Clear ALL

Local database logs are NOT affected — only the data stored on the physical device is cleared.

⚠ **Warning: Clear ALL** permanently removes all users and biometric data from the device. Use **Sync Users** afterward to restore users from another device or the local employee registry.

7. Transfer Data Between Devices

> **Requirement 6:** Manually copy user data (profile, fingerprints, face, photo, card, password) from one device to one or more others — or bidirectionally across all devices. Zone-wise transfer is also supported.

The system supports three sync modes, all accessed from **Manage** → **Sync Users**:

Sync Modes Explained

Mode	Direction	Effect on Target	Use Case
Merge	Host → Targets	Adds/updates users. Extra users on target are left untouched.	Add new users from one device to others
Mirror	Host → Targets	Flushes target first, then copies exactly. Target becomes a copy of host.	Replace all data on a target with a trusted source
Bidirectional	All ↔ All	Every selected device gets every user. No deletions.	Initial setup across multiple new devices

How to Perform a Manual Sync

1. **Click Manage → Sync Users**
2. **Choose a Sync Mode (Merge / Mirror / Bidirectional)**
3. **Select the Host device (source)**
4. **Select one or more Target devices (hold Ctrl/Cmd for multiple)**
5. **Click "Run Sync" — wait for the log to appear**

Sync Users Page	
Sync Mode	( ) Merge ( ) Mirror (●) Bidirectional
Host Device	TC5000-A1B2C3 (conn=0) [not needed for Bidirectional]
Target Devices	☑ TC5000-A1B2C3 (conn=0) ☑ M50-001122 (conn=1) ☑ M91-AABBCC (conn=2)

⟷ Run Bidirectional Merge

Hold Ctrl/Cmd to select multiple targets. Mirror mode will warn before deleting extra users.

Sync Log Output	
	========== BIDIRECTIONAL SYNC: ['TC5000-A1B2C3', 'M50-001122', 'M91-AABBCC'] ========== Pulling from TC5000-A1B2C3... Found 12 user profiles UID=1001 Name=set Fingers=2 Face=Yes UID=1002 Name=set Fingers=3 Face=Yes Pulling from M50-001122... Found 8 users Merged master: 15 unique users M91-AABBCC: Pushing 5 missing users... Push UID=1010: Profile + name + credentials: OK Push UID=1010: Face: OK BIDIRECTIONAL SYNC COMPLETE

Zone-Wise Sync

1. **Go to Manage → Zones / Areas**
2. **Click Sync next to the target zone**
3. **Click Start Zone Sync — all online devices in the zone are synced bidirectionally**

Zone Sync — Ground Floor

Zone	Ground Floor
Devices in Zone	TC5000-ENTRANCE-01, TC5000-EXIT-01, M50-LOBBY
Online Devices Found	3 of 3
Action	Bidirectional merge across all 3 devices

Start Zone Sync

i Data transferred includes: User profile, name, department, access timesets, card number, password, QR code, all 10 fingerprint slots, face template, and user photo.

8. Face / Fingerprint / Card Data — Automatic Sync

> **Requirement 7:** Face, fingerprint, and card data are automatically synchronised across all devices when using the Employee registry or the Sync engine.

Automatic Sync Mechanisms

Trigger	What Happens	Data Included
New employee saved (Manage → Employees)	Profile pushed to ALL currently online devices immediately	Name, card, timesets, period, privilege
Employee edited	Updated profile pushed to ALL online devices	All changed fields
Manual Sync (Manage → Sync Users)	Full biometric sync between selected devices	Profile + fingerprints + face + photo + card
Zone Sync	Bidirectional sync across all devices in a zone	Profile + fingerprints + face + photo + card

How to Verify Automatic Sync

1. Add or edit an employee in **Manage** → **Employees** and click **Save & Push to Devices**.
2. Open any online device → **User Management**.
3. Enter the same Employee ID and click **Read User** — the data should appear on the device.

Employee Push Result	
Action	Save employee EMP-1005 'John Smith'
Devices Online	TC5000-A1B2C3, M50-001122
Push to TC5000-A1B2C3	✔ OK — SetUserData accepted
Push to M50-001122	✔ OK — SetUserData accepted
Biometric sync status	Use Sync Users to push face/fingerprint data

Profile data (name, card, department, timesets) is pushed immediately. Biometric templates (face/fingerprint) require manual sync or enrollment on the device.

Enrolling Biometrics After Auto-Push

After the employee profile is pushed, enroll biometrics remotely (see Section 10) or let the employee enroll directly on the device.

9. Remote Handling of Device Menu

> **Requirement 8:** Enable or disable the device from the software, effectively locking or unlocking the device interface and preventing local operation.

The software allows you to remotely **enable** or **disable** a device's verification interface. When disabled, the device will not process punches or allow enrolment.

How to Enable / Disable a Device

1. **Open the device (Online Devices → Open)**
2. **Click Status / Info / Misc Controls**
3. **Use the Disable Device or Enable Device button**

Status / Info / Misc Controls

Current Time on Device	2026-04-23 10:15:30
Device Status	Door: Closed | Alarm: None
Actions	

Disable Device | Enable Device | Get Time | Set Time (sync) | Get Status | Get Info

Lock Control (Advanced)

For lock/door control, navigate to **Access Control** → **Lock Control** on the device control panel.

Lock Control

Lock Control Mode	[Auto Recover ▼]
Available Modes	Force Open | Force Close | Normal Open | Auto Recover | Restart | Cancel Warning | Illegal Open

Read Mode | Apply Mode

> ℹ **Note: Restart** is available as a dedicated one-click button on the main device control page (see Section 21). All other lock modes are in the Lock Control page.

10. Remote Enrollment — Face, Fingerprint, Card

> **Requirement 9:** Remotely trigger enrollment of face, fingerprint, or card for a specific user ID without physically touching the device.

Method 1 — Remote Enroll (on-device screen)

This method instructs the device to display an enrolment prompt on its own screen so the employee can enrol there.

1. **Device Control Panel → User Data → Remote Enroll**
2. **Enter the User ID and select the enrolment type**
3. **Click Start Enroll — device screen prompts the employee**
4. **Click Query Status to check completion, then Stop Enroll**

Remote Enroll Page

User ID	1005
Enrolment Type	(●) Face ( ) Fingerprint ( ) Card ( ) QR
Finger Slot (if FP)	[Slot 0 ▼] (0–9, left/right hand fingers)
Duplication Check	☑ Yes
Status	Awaiting employee on device screen...

Start Enroll | Query Status | Stop Enroll

Method 2 — Enroll Face by Photo Upload

Upload a JPEG/PNG photo of the employee to enrol their face template remotely.

1. **Device Control Panel → User Data → Enroll Face by Photo**
2. **Enter the User ID and paste the Base64-encoded photo data**
3. **Click Enroll**

Enroll Face by Photo

User ID	1005
Photo Data (Base64)	/9j/4AAQSkZJRgABAQAAAQABAAD/... [truncated]

Enroll Face

The device extracts the face feature vector from the photo and stores the template. Best results with a clear front-facing photo.

Managing Biometric Data Directly

Use the dedicated pages under **User Data** on the device control panel:

- **Manage Face Data** — Read / Write / Delete face templates
- **Manage Fingerprints** — Read / Write / Delete any of the 10 fingerprint slots
- **View Password / Card / QR** — Read card number, QR code, password from a user

11. Device Connect / Disconnect Logging

> **Requirement 10:** Every time a device connects to or disconnects from the system, the event is logged with a timestamp and stored in the database.

How Connection Events are Logged

When a device logs in successfully, the Device Broker posts a `connect` event to Django. When the WebSocket connection closes, a `disconnect` event is posted. Both are stored in the **DeviceConnectionLog** table.

How to View Connection Logs

1. **Click Manage → Connection Logs in the top navigation bar**

Device Connection Logs

Timestamp	Device ID	Event
2026-04-23 10:15:02	TC5000-A1B2C3	🟢 Connected
2026-04-23 09:58:11	M50-001122	🔴 Disconnected
2026-04-23 09:58:00	M50-001122	🟢 Connected
2026-04-23 08:30:45	M91-AABBCC	🟢 Connected
2026-04-22 23:01:10	TC5000-A1B2C3	🔴 Disconnected

The 200 most recent events are shown. Green = Connected, Grey = Disconnected.

> **i Where stored:** The `DeviceConnectionLog` table in the database. The `DeviceRegistry` record for each device is also updated with the `last_seen` timestamp on each connect.

12. Time Zone Access per Employee

Requirement 11: Each employee can be assigned up to 5 access time zones that define which hours of each day of the week they are allowed to access a device.

Step 1 — Define an Access Timezone Schedule

1. **Open a device → Access Control → Access Timezone Settings**
2. **Enter a Timezone Number (1–50) and define the time sections for each day**
3. **Click Write Settings**

Access Timezone Settings

Timezone Number	1
Sunday (Section 0)	Start: 0 End: 0 (= no access)
Monday (Section 1)	Start: 480 End: 1080 (08:00 – 18:00)
Tuesday (Section 2)	Start: 480 End: 1080
Wednesday (Section 3)	Start: 480 End: 1080
Thursday (Section 4)	Start: 480 End: 1080
Friday (Section 5)	Start: 480 End: 1080
Saturday (Section 6)	Start: 0 End: 0 (= no access)

Read Settings | Write Settings

Times are in minutes since midnight. 480 = 08:00, 1080 = 18:00. 0,0 means no access that day.

Step 2 — Assign Timezone to an Employee

1. **Open a device → User Management → enter User ID → Read User**
2. **Set Timeset 1 to the timezone number (e.g. 1) and leave others at -1**
3. **Click Write User**

User Management — Timeset Assignment

User ID	1005
Name	John Smith
Timeset 1	1 ← timezone #1 (Mon-Fri 08:00-18:00)
Timeset 2	-1 (not set)
Timeset 3	-1
Timeset 4	-1
Timeset 5	-1

Read User | Write User

Or assign via **Manage** → **Employees** → **Edit Employee** → **Timeset 1–5** fields — this pushes to all devices automatically.

13. Add and Delete Employees

Requirement 12: Add new employees or delete existing ones from the software. Changes are automatically pushed to all connected devices.

Adding an Employee

1. **Click Manage → Employees in the top navigation bar**
2. **Click + Add Employee**
3. **Fill in the employee details**

Add Employee Form

Employee ID *	1006
Name *	Priya Sharma
Department	3
Privilege	[User ▼] (User / Manager / Administrator)
Enabled	☑ Yes
Card Number	A2B3C4D5
Password	1234
Period Start (block start)	2026-01-01
Period End (block end)	2026-12-31
Timeset 1	1
Timeset 2–5	-1

Save & Push to Devices Cancel

Clicking 'Save & Push to Devices' immediately sends this employee's profile to ALL currently online devices.

4. **Click Save & Push to Devices**

Employees List

ID	Name	Dept	Privilege	Enabled	Card	Period	Actions
1001	Ahmed Ali	1	User	🟢 Yes	A1B2C3D4	—	Edit | Disable | Delete
1005	John Smith	2	Manager	🟢 Yes	—	2026-01-01 → 2026-12-31	Edit | Disable | Delete
1006	Priya Sharma	3	User	🟢 Yes	A2B3C4D5	2026-01-01 → 2026-12-31	Edit | Disable | Delete

Deleting an Employee

1. **Click Delete next to the employee in the list**
2. **Confirm the deletion on the confirmation page**

Delete Employee Confirmation

Employee	1006 — Priya Sharma
Warning	This will delete the employee from the software AND from ALL connected devices.

Delete from Software & All Devices | Cancel

The delete command (SetUserData Type=Delete) is sent to every online device before removing the database record.

Deleting from a Single Device (Device-Level)

To delete a user from only one specific device (not from the employee registry), use the per-device **User Management** page:

1. Open the device → User Management
2. Enter the User ID and click **Read User**
3. Click **Delete User**

14. Delete Logs by Period

Requirement 13: Delete attendance or management logs for a specific date/time range from the local database.

Method 1 — Delete DB Logs by Date Range

1. **Click Logs → Attendance Logs in the top navigation bar**
2. **Fill in the filter fields: Device ID, User ID, Start Time, End Time**
3. **Click Clear Filtered Logs to delete only the matching records**

Attendance Logs — Delete by Period

Device ID (optional)	TC5000-A1B2C3
User ID (optional)	1005
Start Time	2026-01-01 00:00
End Time	2026-03-31 23:59
Matched Records	243 logs found

Search | Clear Filtered Logs

Only logs in the local database are affected. Logs on the device itself are not deleted by this action.

Method 2 — Delete Logs from the Physical Device

1. **Open device → View Logs (On-Demand Log Fetching) → Attendance Logs**
2. **Click Get Log Pos Info to see the log position range**
3. **Enter the log position ID up to which you want to delete and click Delete Logs**

View Device Attendance Logs

Log Count on Device	1842
Start Position	0
Max Count	100000
Next Log ID (delete up to)	500

Get First Log | Get Next Log | Get Log Pos Info | Delete Logs

⚠ On-device deletion removes logs up to a position index, not by date. To delete by date, first fetch logs with a date filter to find the position, then delete up to that position.

Delete Management Logs by Period

Follow the same steps using **Logs** → **Management Logs** → filter by date range → **Clear Filtered Logs**.

15. Download Logs — Automatic and Manual

> **Requirement 14 & 15:** Attendance logs are downloaded automatically when the device pushes them, and can also be fetched manually with a from/to date range. Both new (real-time) and old (historical) logs are supported.

Automatic Log Download

Every time an employee punches in/out on a device, the device immediately pushes the log to the server over the existing WebSocket connection. No manual action is required.

Automatic Log Flow	
Step 1	Employee punches on device
Step 2	Device sends TimeLog XML event to broker
Step 3	Broker posts to Django /device/upload_log
Step 4	Django saves to AttendanceLog table
Step 5	Log visible in Logs → Attendance Logs within seconds

Manual Log Download — Bulk (with Date Range)

To fetch historical logs stored on a device for a specific date range:

1. **Open the device control panel → Maintenance → Bulk Download Logs**
2. **Set the Start and End date/time**
3. **Click Download**

Bulk Download Logs from Device	
Start Date/Time	[2026-01-01T00:00]
End Date/Time	[2026-03-31T23:59]
Result	✔ Saved 1,842 log(s) to the database.

Download

Leave both fields blank to download ALL logs from the device. Duplicate logs (same device_id + log_id) are skipped automatically.

Manual Log Download — Record by Record (On-Device Fetch)

1. **Open device → View Logs (On-Demand) → Attendance Logs**
2. **Optionally set User ID, Start Time, End Time filter**
3. **Click Get First Log, then Get Next Log repeatedly**

Export Logs to CSV

1. **Click Logs → Export CSV in the top navigation bar**
2. **Optionally add URL filters:** `?device_id=TC5000-A1B2C3&start=2026-01-01&end=2026-03-31`
3. **The browser downloads a CSV file**

CSV Export URL Examples

All logs	/logs/download_csv/
By device	/logs/download_csv/?device_id=TC5000-A1B2C3
By date range	/logs/download_csv/?start=2026-01-01&end=2026-03-31
Combined	/logs/download_csv/?device_id=TC5000-A1B2C3&start=2026-01-01&end=2026-03-31

CSV columns: ID, Device, LogID, Time, UserID, AttendStatus, Action, AttendOnly, Expired

16. Time Synchronisation

Requirement 16: Synchronise the device clock with the server's current date and time, or configure an NTP server for automatic time sync.

Method 1 — One-Click Time Sync (Set to Server Time)

1. **Open device → Status / Info / Misc Controls**
2. **Click Set Time (Sync) — the device clock is immediately set to the server's current time**

Time Sync

Current Time on Device	2026-04-23 09:45:10
Server Time	2026-04-23 10:15:30
Time Difference	30 minutes 20 seconds behind

Get Device Time | Set Time (Sync to Server)

After clicking 'Set Time', the device clock is updated to match the server's UTC clock instantly.

Method 2 — NTP Server Configuration

1. **Open device → NTP Server Settings**
2. **Enter the NTP server address, UTC offset, and sync interval**
3. **Click Write Settings**

NTP Server Settings

NTP Server Address	pool.ntp.org
UTC Timezone Offset	+330 (= UTC+5:30 for IST)
Sync Interval (min)	60

Read Settings | Write Settings

UTC offset is in minutes. Examples: +330 = IST (India), +480 = CST (China), 0 = UTC, -300 = EST.

17. Enable and Disable Users

Requirement 17: Enable or disable a specific user across all devices or on a single device. A disabled user cannot punch in.

Method 1 — Toggle from Employee List (All Devices)

1. **Click Manage → Employees**
2. **Click the Disable or Enable button next to the employee**
3. **The system immediately pushes the updated status to all connected devices**

Employees List — Enable/Disable

ID	Name	Enabled	Actions
1001	Ahmed Ali	🟢 Yes	Edit | **Disable** | Delete
1005	John Smith	🔴 No	Edit | **Enable** | Delete
1006	Priya Sharma	🟢 Yes	Edit | **Disable** | Delete

Disabled users appear greyed out. Clicking Disable/Enable immediately updates all connected devices.

Method 2 — Enable/Disable on a Single Device

1. **Open device → User Management → enter User ID → Read User**
2. **Change the Enabled field to Yes or No**
3. **Click Write User**

User Management — Enable/Disable

User ID	1005
Name	John Smith
Enabled	( ) Yes (●) No ← disabled on THIS device only

Write User

This only affects the selected device. Use the Employee list method to push to all devices simultaneously.

18. Name Upload to Device

Requirement 18: Upload an employee's name (and other profile data) to one or more devices.

Employee names are always included when writing user data. The name is encoded in UTF-16LE Base64 format as required by the device firmware.

How Names are Uploaded

Action	How Name is Sent
Add Employee (Manage → Employees)	Name pushed to all online devices automatically
Edit Employee	Name (re-)pushed to all online devices
User Management → Write User (per device)	Name sent to that specific device
Sync Users	Name transferred to all target devices

Upload Name via User Management (Single Device)

1. **Open device → User Management**
2. **Enter the User ID and Name**
3. **Click Write User**

User Management — Name Upload

User ID	1006
Name	Priya Sharma
Privilege	User
Enabled	Yes
Department	3

Write User

The name is transmitted in UTF-16LE encoding to support all languages including Arabic, Chinese, Hindi, etc.

19. Auto-Upload New Employee to All Devices

> **Requirement 19:** When a new employee is registered in the software, their profile is automatically pushed to all currently connected devices without any manual action.

How It Works

When you save a new employee using **Manage** → **Employees** → **Add Employee**, the system:

1. Saves the employee record to the local database.
2. Immediately opens a connection to the Device Broker.
3. Calls `SetUserData` on every currently online device.
4. The employee is available on all devices within seconds.

Auto-Upload Flow

1. User clicks	Save & Push to Devices
2. DB record created	Employee ID=1007 'Sara Khan' saved
3. Devices online	TC5000-A1B2C3 (conn=0), M50-001122 (conn=1), M91-AABBCC (conn=2)
4. Push TC5000	✔ SetUserData → OK
4. Push M50	✔ SetUserData → OK
4. Push M91	✔ SetUserData → OK
5. Result	Employee available on 3 devices

If a device is offline when the employee is added, use Sync Users later to push to that device.

> **i What is pushed automatically:** Name, privilege, enabled status, department, card number, password, 5 timeset slots, and validity period.
> **What requires manual sync:** Fingerprint templates and face templates (must be enrolled via Remote Enroll or Sync Users).

20. Date / Time Employee Block

> **Requirement 20:** Set a validity period for an employee so that the device automatically denies access before the start date or after the end date.

Every employee record has a **Period Start** and **Period End** date. When these are set, the device only allows the employee to punch in during that date window.

Set a Validity Period for an Employee

1. **Go to Manage → Employees and click Edit for the employee**
2. **Set Period Start and Period End dates**
3. **Click Save & Push to Devices**

Employee Form — Period Block

Employee ID	1008
Name	Contract Worker A
Period Start	2026-05-01 ← access allowed from this date
Period End	2026-07-31 ← access blocked after this date
Result on Device	Device rejects punch attempts outside 1 May – 31 July 2026

Save & Push to Devices

To block immediately: set Period End to today's date. To unblock: clear both fields.

Set Period via Single-Device User Management

1. **Open device → User Management → Read User**
2. **Fill in Period Start and Period End fields**
3. **Click Write User**

i Combined with Access Timezones (Section 12): Use the Period to control *which dates* an employee has access, and Access Timezones to control *which hours* of each day.

21. Restart Device from Software

Requirement 21: Remotely restart a biometric device without physical access.

Method 1 — Dedicated Restart Button (One Click)

1. **Open the device control panel (Online Devices → Open)**
2. **Scroll to Maintenance section at the bottom of the page**
3. **Click the Restart Device button**
4. **Confirm the restart dialog**

Device Control Panel — Maintenance Section	
Clear Data	→ Open Clear Data page
Firmware Version	→ Check current firmware
Write Firmware	→ OTA firmware update
Bulk Download Logs	→ Download logs with date range

Clear Data | Firmware Version | Write Firmware | Bulk Download Logs | Restart Device

Clicking 'Restart Device' shows a browser confirmation dialog. On confirm, the device reboots within 3-5 seconds.

Restart Confirmation Response	
Result	{ "ok": true }
Device Status	Offline (rebooting) → Online (connected) within ~30 seconds

Method 2 — Via Lock Control (Manual Mode)

1. **Open device → Access Control → Lock Control**
2. **Select Restart from the Mode dropdown**
3. **Click Apply Mode**

✓ **After restart:** The device reconnects automatically within 30–60 seconds. A new Connected entry appears in the Device Connection Logs.

22. Interlock

> **Requirement 22:** When an employee punches on one interlock-enabled device, access is automatically denied (user disabled) on all other interlock-enabled devices until the next valid punch.

How Interlock Works

The Interlock feature prevents "tailgating" — an employee cannot badge into two different doors simultaneously. The logic is:

1. Employee punches on Device A (interlock enabled).
2. The punch event reaches Django via the log upload endpoint.
3. Django finds all other online interlock-enabled devices.
4. Django returns their connection IDs to the broker.
5. The broker sends `SetUserData Enabled=No` to those devices for that employee.
6. The employee cannot punch on any other interlock device until they punch again on a valid device.

Interlock Flow Diagram	
1. Punch	EMP-1005 punches on TC5000-ENTRANCE (interlock=ON)
2. Log uploaded	Django receives TimeLog for EMP-1005
3. Interlock check	TC5000-ENTRANCE has interlock=ON → trigger
4. Find others	M50-EXIT (conn=1), M91-CAFETERIA (conn=2) — also interlock=ON
5. Deny on M50	Broker sends SetUserData UserID=1005 Enabled=No → M50
6. Deny on M91	Broker sends SetUserData UserID=1005 Enabled=No → M91
7. Result	EMP-1005 can only exit through the punch sequence

Step 1 — Enable Interlock on Devices

1. **Go to Manage → Device Registry**
2. **Click Edit for each device that should participate in interlock**
3. **Check Enable Interlock and click Save**

Device Registry Edit — Enable Interlock

Serial Number	TC5000-ENTRANCE-01
Friendly Name	Main Entrance Reader
Interlock Enabled	☑ Enable Interlock (deny access on other interlock devices after punch)

Save

Step 2 — Alternatively, Use the Interlock Status Page

1. **Click Manage → Interlock Status**
2. **Click Enable or Disable next to each device**

Interlock Status Page

Device	Zone	Interlock	Action
TC5000-ENTRANCE-01 (Main Entrance)	Ground Floor	🟡 ON	Disable
M50-EXIT-01 (Exit Door)	Ground Floor	🟡 ON	Disable
M91-CAFETERIA (Cafeteria)	1st Floor	⚪ OFF	Enable

Step 3 — View Recent Interlock Events

Interlock Events Log

Employee ID	Punched Device	Punch Time	Status
1005	TC5000-ENTRANCE-01	2026-04-23 09:02:15	🟡 Active
1001	M50-EXIT-01	2026-04-23 08:58:44	⚪ Cleared
1006	TC5000-ENTRANCE-01	2026-04-23 08:45:10	⚪ Cleared

⚠ **Important:** Interlock works in real time as punches occur. For it to function, the Device Broker and Django must be running and the devices must be online.

23. API Integration for External Software

> **Requirement 23:** A secure REST API allows HR systems, payroll software, ERP platforms, and other external applications to query and manage the device management system programmatically.

Authentication

All API endpoints require an API key passed in the `X-API-Key` HTTP header (or as a `?api_key=` URL parameter).

Step 1 — Generate an API Key

1. **Click API Keys in the top navigation bar**
2. **Enter a name for the key (e.g., "HR Software") and click Generate New Key**

API Key Management

Name	Key	Status	Last Used	Actions
HR Software	a1b2c3d4e5f6...32chars	🟢 Active	2026-04-23 10:05	Disable | Delete
Payroll System	9f8e7d6c5b4a...32chars	🟢 Active	Never	Disable | Delete

REST API Endpoints

```
GET /api/v1/devices/
→ List all currently online devices

GET /api/v1/employees/
→ List all employees in the database

POST /api/v1/employees/
→ Create a new employee and push to all online devices

GET /api/v1/employees/<employee_id>/
→ Get details for one employee

PUT /api/v1/employees/<employee_id>/
→ Update an employee and push changes to all devices

DELETE /api/v1/employees/<employee_id>/
→ Delete employee from DB and all connected devices
```

```
GET /api/v1/logs/?device_id=&user_id=&start=&end=
→ Query attendance logs (up to 500 records)

POST /api/v1/sync/
→ Trigger a device sync (merge / mirror / bidirectional)
```

Example: Create an Employee via API

```
curl -X POST https://your-server/api/v1/employees/ \
  -H "X-API-Key: a1b2c3d4e5f6..." \
  -H "Content-Type: application/json" \
  -d '{"employee_id": 2001, "name": "Ali Hassan", "department": 5, "card":
"FFAA1122", "enabled": true}'

Response: 201 Created
{"id": 42, "employee_id": 2001}
```

Example: Query Logs

```
curl "https://your-server/api/v1/logs/?device_id=TC5000-
A1B2C3&start=2026-04-01&end=2026-04-23" \
  -H "X-API-Key: a1b2c3d4e5f6..."

Response: 200 OK
{"logs": [{"id": 101, "device_id": "TC5000-A1B2C3", "time": "2026-04-23T09:02:15",
"user_id": 1005, "attend_status": "CheckIn", ...}]}
```

Example: Trigger Sync

```
curl -X POST https://your-server/api/v1/sync/ \
  -H "X-API-Key: a1b2c3d4e5f6..." \
  -H "Content-Type: application/json" \
  -d '{"host": "TC5000-A1B2C3", "targets": ["M50-001122", "M91-AABBCC"], "mode":
"merge"}'

Response: 200 OK — includes full sync log array
```

API Parameter Reference

Parameter	Type	Description
`employee_id`	integer	Unique employee number (matches User ID on device)
`name`	string	Full name (UTF-8, supports all scripts)

Parameter	Type	Description
`department`	integer	Department number (0–99)
`privilege`	integer	0=User, 1=Manager, 2=Administrator
`enabled`	boolean	true = can punch; false = denied on all devices
`card`	string	Card / RFID number (hex or decimal)
`password`	string	Door password
`timeset_1–5`	integer	Access timezone slot numbers (-1 = not set)
`period_start / period_end`	date (YYYY-MM-DD)	Validity window — device denies outside range
`mode (sync)`	string	"merge" | "mirror" | "bidirectional"

✓ **Integration examples:** HR software can POST new hires on their first day. Payroll can GET logs for the month. Security systems can PUT employee.enabled=false on termination — all devices update within seconds.

Appendix — Quick Reference

Key URLs

URL	Purpose
`/`	Home / Dashboard
`/online_devices`	Live device list
`/device_registry/`	Persistent device management
`/zones/`	Zone / Area management
`/employees/`	Employee registry (auto-push)
`/sync_users/`	Manual user sync between devices
`/interlock/`	Interlock configuration & events
`/device_connection_logs/`	Connect/disconnect log
`/search_attend_logs`	Attendance log viewer & filter
`/logs/download_csv/`	Export attendance logs as CSV
`/api_keys/`	REST API key management
`/api/v1/employees/`	REST API — employee CRUD
`/api/v1/logs/`	REST API — log query
`/api/v1/devices/`	REST API — online device list
`/api/v1/sync/`	REST API — trigger sync
`/control_device/<id>/restart/`	Restart device (AJAX)
`/control_device/<id>/bulk_download_logs/`	Bulk log download with date range

Troubleshooting

Problem	Likely Cause	Fix
Device not appearing in Online Devices	Wrong server URL on device, or broker not running	Check broker is running; verify device server URL setting
Login rejected (token invalid)	Token mismatch or device not in registry	Go to Device Registry, find/add device, reset token if needed
Sync fails with "Host not found"	Host device offline	Ensure host device is in Online Devices list before syncing

Problem	Likely Cause	Fix
Employee push fails silently	No devices online	Check Online Devices — at least one must be connected
Interlock not triggering	Device not in DeviceRegistry with interlock_enabled=True	Go to Device Registry → Edit → Enable Interlock
API returns 401 Unauthorized	Missing or invalid API key	Generate key at /api_keys/ and pass as X-API-Key header
CSV export empty	No logs match the filters	Check date range and device_id parameters